PeriGen
Advanced Perinatal Systems

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PeriWatch® Tracings™
Interface Specification

1 Single Sign On

Clinicians charting in their EHR system often find use for quickly viewing the maternal-fetal surveillance strip, as well as viewing or modifying annotations to this strip. To that end, PeriWatch Tracings (PWT) has developed a single sign-on (SSO) integration whereby the user of an outside charting system may launch a special PWT surveillance screen without need of authenticating to PWT. This process and the enabling mechanism are here described.

1.1 Design Detail

Users of an EHR wishing to launch the PWT SSO integration will be provided a launch link, button, or other mechanism by this host system. The host system will launch an executable, the Message Dispatcher in one of two modes

1.1.1 Command Line Invocation

One time use where the launch information is sent via command line. The Message Dispatcher then calls the Launch Service to pass the message along. In this care there will be a launch per use of SSO launch. The command-line arguments contain the context and authentication required to successfully launch a relevant session of PWT.

1.1.2 File Watching Invocation

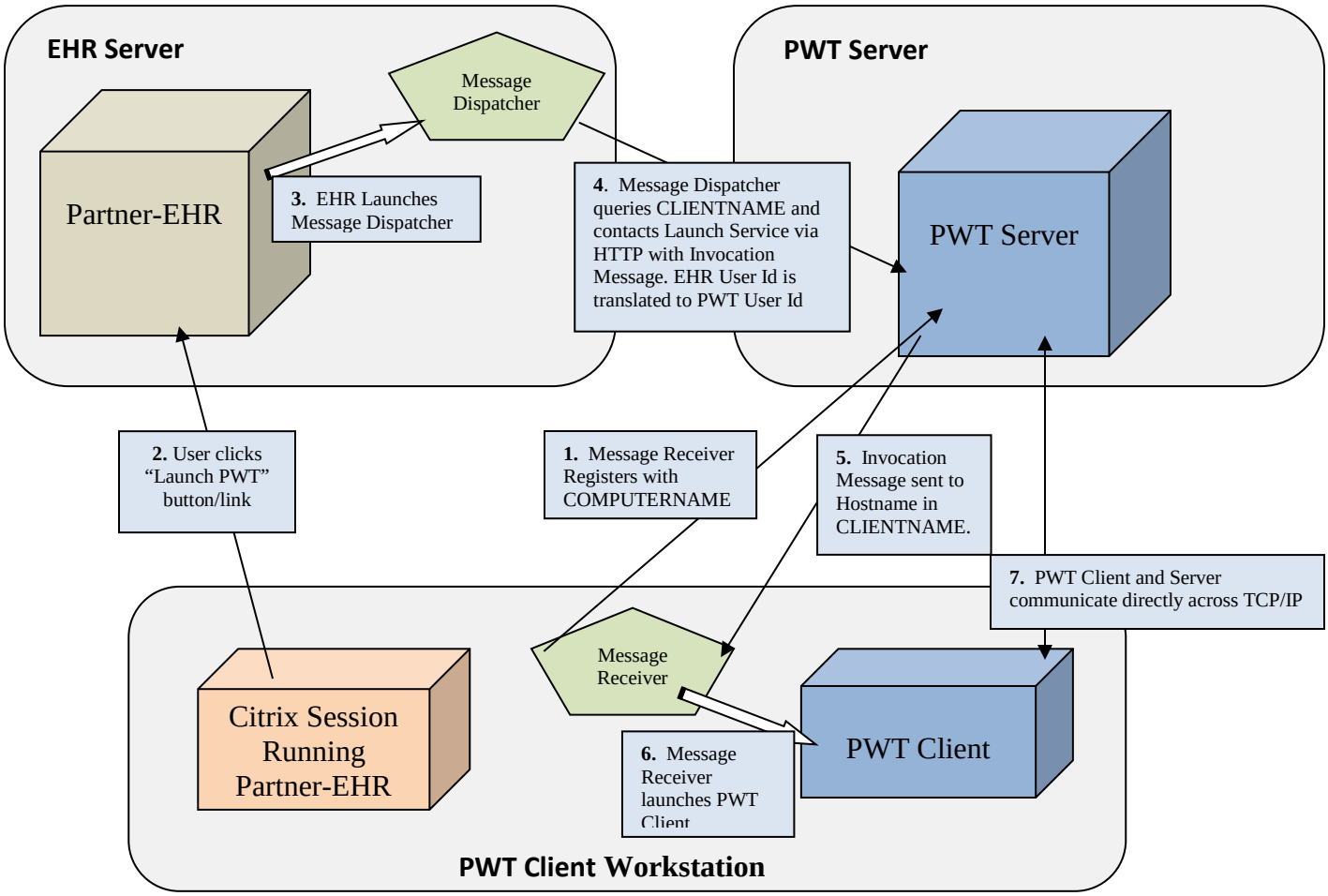
Beginning with PeriWatch 2017.Rev3 support for a new file-watch SSO mode has been introduced, where the EHR can drop a file in a predesignated area indicating the user actions like Login, Logout and Patient Switch. These actions will be translated into appropriate PeriWatch launch messages and sent to the Launch Service The files contain the context and authentication, in a predefined format, required to successfully launch a relevant session of PWT. This support is currently for Meditech based EHR.

The Message Dispatcher resides on and is invoked from a central server, and will communicate with a *Launch Service* on the PWT Server, The PWT server informs a resident and always-running Message Receiver on the user's client workstation. The Message Receiver already has registered with the PWT server using the value in the environment variable COMPUTERNAME

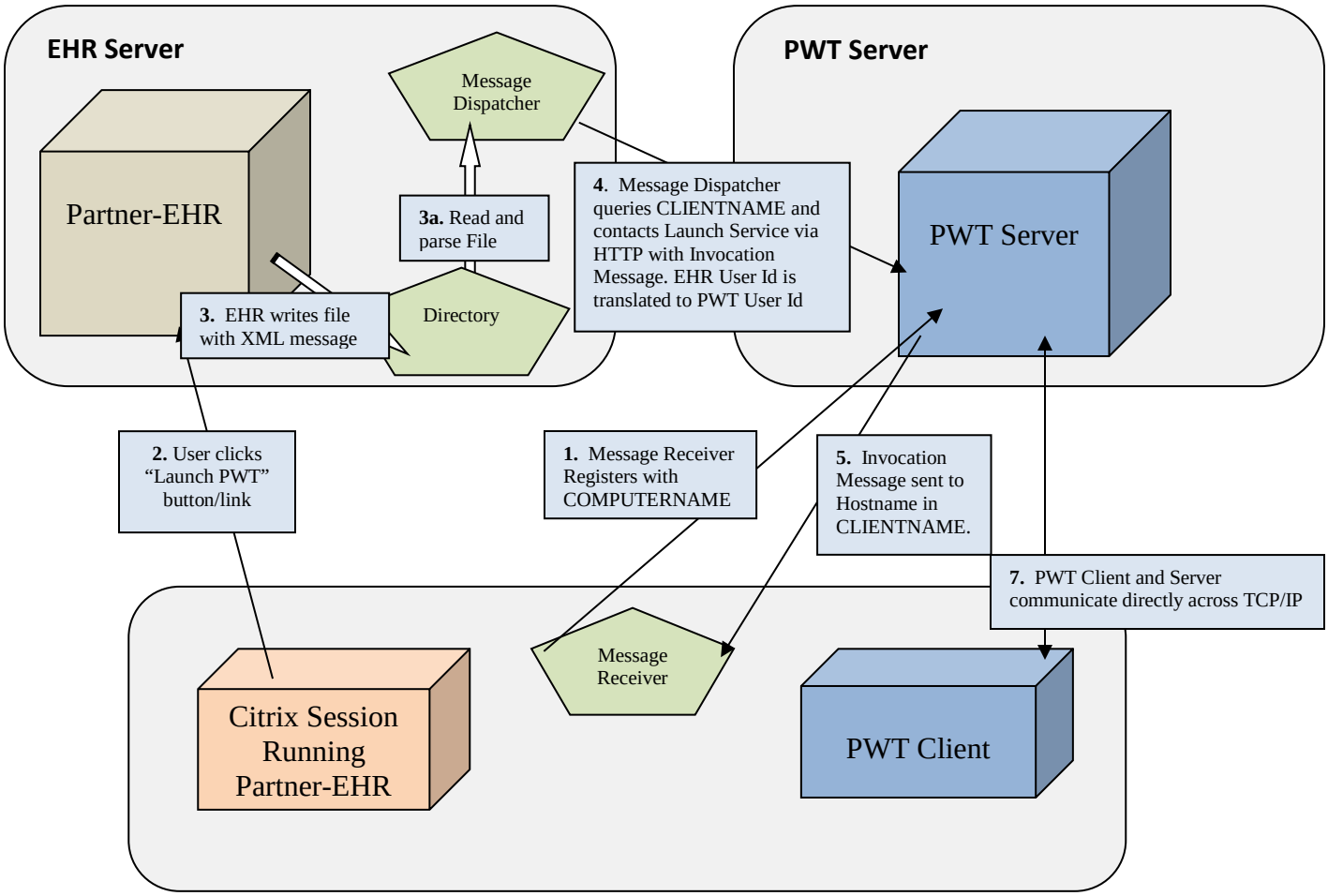
The Message Receiver, upon receiving the encoded instructions, will launch PWT on the client workstation in a special windowed-session, affect a login, and automatically display the surveillance strip of the patient indicated by the EHR at time of launch.

The figures below illustrate the two modes. As can be noticed the modes are almost identical except for the behavior of the Message Dispatcher.

1.2 Command Line Invocation Mode



1.3 File Watching Invocation Mode



Implementation of this workflow will require access to an open port on the client workstation, defaulting to 8009 but configurable, to traffic from the PWT server. This port is only used for communication with the PWT server. In addition, access to HTTP port of the PWT server is also required.

The mode being used and the file location to watch in the case of the File Watching mode all require configuration changes to the Launch Service that would require PeriWatch Operations or Technical support

1.4 User Mapping

PWT user names may differ from those of the partner-EHR. To accommodate such configuration, PWT provides a real-time translation service (step four above), whereby EHR userIDs are translated during the process of SSO launch. This service also applies to interfacing of, e.g., annotations inbound to PWT from the partner-EHR.

Users at configured sites will need to provide the PWT Integrations Team representative with a list of users authorized to use the SSO feature. Typically, this takes the form of an Excel spreadsheet with two columns:

EHRuserID1	PWTuserID1
EHRuserID2	PWTuserID1
EHRuserID3	PWTuserID2

Note that multiple EHR userIDs may be mapped to a single PWT userID. Upon receipt of a launch request (or interface message) containing EHRuserIDx, PWT agents will query a database table for the corresponding PWT userID and perform an inline substitution.

Maintenance of this lookup table requires submission of new userID pairs to the PWT Integrations Team. There is a tool planned for development which will allow direct user creation/modification of this mapping; stay tuned.

1.5 Implementation Detail Message Dispatcher setup (Pre-PeriWatch 2017 Rev3)

1.5.1 Configure the Message Dispatcher

It is best to acquire the Message Dispatcher folder from the PWT Server, since it is appropriately configured to communicate with the PWT Server. In case that is not feasible then configure of the SSO function is primarily via changes made to the MessageDispatcher.exe.config by changing important site-specific keys, however the four most commonly modified keys are:

```
<add key="IPAddress" value="127.0.0.1"/>
<add key="TCPPort" value="8009"/>
<endpoint address=http://[PWT Server IP]/AuthenticationService.svc… />
<endpoint address=http://[PWT Server IP]/UserDisambiguationService.svc… />
```

These addresses are to be changed to reference the appropriate PWT server at a given installation site; the port may be modified at client request.

1.5.2 Configure the EHR software

The partner-EHR will need to provide users with a means of launching PWT. This launch mechanism must provide context to PWT – transmitting the First and Last Name, DOB, Medical Record Number, and Visit Number of the patient of interest – as well as authentication credentials in the form of a user ID and authentication string. The message dispatcher is to be invoked on each launch of PWT with the following command-line argument string:

--url Auth=123456^XXXXXX&DOB=19800505&MRN=654321&PatientName=Last^First^M&UserID=UID

…where	123456	=	Patient's Visit Number
	XXXXXX	=	Authentication String (encoded/hashed value)
	19800505	=	Patient's Date of Birth
	654321	=	Patient's Medical Record Number
	Last^First^	=	Patient's Name in Last, First, MI format, and
	UID	=	UserID of user to be signed-on in PWT.

1.6 Implementation Detail Message Dispatcher setup (Since PeriWatch 2017 Rev3)

1) Obtain a copy of the Message Dispatcher files from a configured PWT Server. A server that has already been upgraded and setup for operation of the PeriWatch system.
2) For the command line SSO implementation nothing further must be done to the Message Dispatcher, other than having it available for launch by the EHR software when a button is clicked in the EHR software by the user.
3) For the file based system there are more configurations to be done on the PWT server side for issues like
    a. The location of the directory where the files will be deposited by the EHR software
    b. Whether there will be one instance of Message Dispatcher software running for the entire EHR application server or not. This decision is made necessary because with the file based mode the Message Dispatcher must run continuously to react to file creation.
    c. The File based approach requires that the hostname be available so that messages can be embedded. In the case that there is only one instance of Message Dispatcher running for multiple EHR sessions, the hostname would have to be communicated via either the filename or embedded in the directory name that is being watched (perhaps one directory per hostname).

After all these decisions are made PeriGen operations and Tech Support will set up the Launch Service configurations appropriately in a customize configuration file residing on the PeriWatch Server. The Message Dispatcher has been modified to request and use this configuration and set itself up appropriately.

1.7 Installation/Configuration Detail

Installation of the SSO function currently requires a version upgrade with downtime. At such a time:

- The database is modified to contain the appropriate tables and content for the user cross-mapping
- The message receiver is pushed to client workstations,
- A registry key providing for the future startup of Message Receiver must be inserted on each workstation,
- Pre-PeriWatch 2017 Rev3, customers should copy the Message Dispatcher folder to appropriate EHR host server(s) and configure their host EHR launch button/link to invoke this executable with the previously-described, properly-formatted command-line argument

- Since PeriWatch 2017 Rev3 customers, coordinating with PeriGen Operations need to agree
    - On the location of the folders where files will be deposited to enable the Message Dispatcher to be able to read them.
    - This location will then be configured into the Launch service settings on the PeriWatch Server (See Section 1.6)
- The PWT integrations engine configuration is updated to enable inbound annotations.

Additionally, Pre-PeriWatch 2017 Rev 3 an authentication and user-disambiguation service are installed on the PWT server, to be invoked and/or accessed by the message dispatcher on SSO launch. Since 2017 Rev3 this is not needed as installation is automatic.

1.8 Current Integrations

- CommandLine Invocation Mode is currently used with the Epic, AllScripts and Cerner EMRs
- With PWT 2017.Rev3, File Watch Invocation Mode is now supported with Meditech 6.1.5

2 Maternal Interface Specification

The purpose of this section is to outline the PeriWatch Tracings (**PWT**) expectations for HL7 message exchange with other systems. This document should form the foundation for interface design discussions, and presumes prior knowledge of the Health Level Seven (**HL7**) messaging standard; details available at http://www.hl7.org.

2.1 Communications

The PWT integrations engine resides on PWT training and production application servers, there sharing server IPs, network domain authorization/restrictions, and server hardware and database with the related application. As hardware and database are shared between application and integrations engine, PWT expresses strong preference towards the configuration of a **client-side filter**, where possible only passing messages related to existing or potential PWT patients to our engine for consideration and consumption. Often this filtering is accomplished by logic parsing PV1.3.1, passing messages only for patients with an assigned location of (e.g.) LND, LDRP, and/or OP OB Clinic; specific implementation of these filters is at client discretion. Additional requisite filtering may be required; details below where relevant.

Interface messaging occurs across TCP/IP connections. Our standard implementation port for incoming messages is **17676**. Outbound message destinations are expected to be the IP and port of a receiving interface engine; however, data may also be placed on a local file structure in various formats for collection at client preference.

Splitting of inbound message types occurs within the engine, and therefore messages should be collected at client systems and transmitted to PWT in a unitary stream where multiple system are involved (e.g., collect ADT, Lab, and Pharmacy messages for serial transmission to PWT on port 17676). Due to the loosely-coupled nature of the integrations engine with the application, messages need not be properly sequenced to be properly processed (e.g., reception of a lab message for a patient for which we have not yet received an ADT message is acceptable; reception of an ADT update message prior to reception of an ADT admit message is acceptable).

PWT is configured for serialized/bimodal message handling, transmitting or waiting for reception of acknowledgment messages immediately after message reception. Failed transmissions will retry or wait for retry until five attempts have passed, then closing the port and marking the route in error.

The low-level format of the HL7 data stream adheres to standard:

<SB>dddd**<EB><CR>**

…where **<SB>** is the start block character (1 byte ASCII <VT>, 0x0B),
dddd represents the HL7 message data stream,
<EB> is the end block character (1 byte ASCII <FS>, 0x1C), and
<CR> is the carriage return character (0x0D).

2.2 Charting Interface Description

Where a typical downstream ancillary will react to the reception of an ADT^A01 message by admitting that patient to their census, PWT will not take any action – instead the message contents are placed in a pool of data from which a user

can draw information during a manual admission into the application. Similarly, pharmacy inbound messages are primarily for pre-population of medication information when users document an administration in our application. As this is a deviation from common expectation, feel free to ask for more detail.

HL7 Message Definitions

The PWT interface standard adheres to the published HL7 standards for any v2.x message type. Valid transmitted message segments which are not mentioned in the following message definitions are ignored by our application.

Message delimiters adhere to standard: |,^,&,~, and \ being reserved characters to delimit fields, components, subcomponents, repeating fields, and escaped characters (respectively).

PWT currently has the capacity to process inbound HL7 messages pertaining to registration (ADT), lab results (ORU), and Pharmacy messages (RDS, RDE, and RAS). Inbound data from host EHR systems are accepted as specially-formatted ORU unsolicited result (R01) messages and/or inside OBX message segments appended to ADT messages. PWT also has facilities for the export of vitals and annotations, as well as metadata surrounding discharge and audit reporting. Provision of additional inbound or outbound message exchange is outside the scope of this document.

The following sections define the required and optional components of PWT HL7 messages. Indentations denote logical message sub-blocks; fields in curly braces (**{}**) may repeat, those in square brackets (**[]**) are optional per specification.

Message Structures

ADT^A0{1,2,3,4,5,8}	Admit, Registration, PreAdmit, Update
MSH	Message Header
EVN	Event Type
PID	Patient Identifier
NK1	Next-of-Kin / Emergency Contact
PV1	Patient Visit Information
[{ **PV2** }]	Additional Patient Visit Information
[{ **OBX** }]	Observation Segments
[{ **AL1** }]	Patient Allergy Information
[{ **DG1** }]	Patient Diagnoses

Admit, Discharge, Transfer (ADT)

PWT only processes received Admit, Transfers, Discharges, Registration, Pre-Admit, and Update (A01, A02, A03, A04, A05, and A08, respectively) messages; all others are ignored and should be filtered from transmission. PWT does not transmit any ADT messages.

ADT Example

The following section describes the expected behavior of a “standard” sample of ADT and Lab HL7 messages.

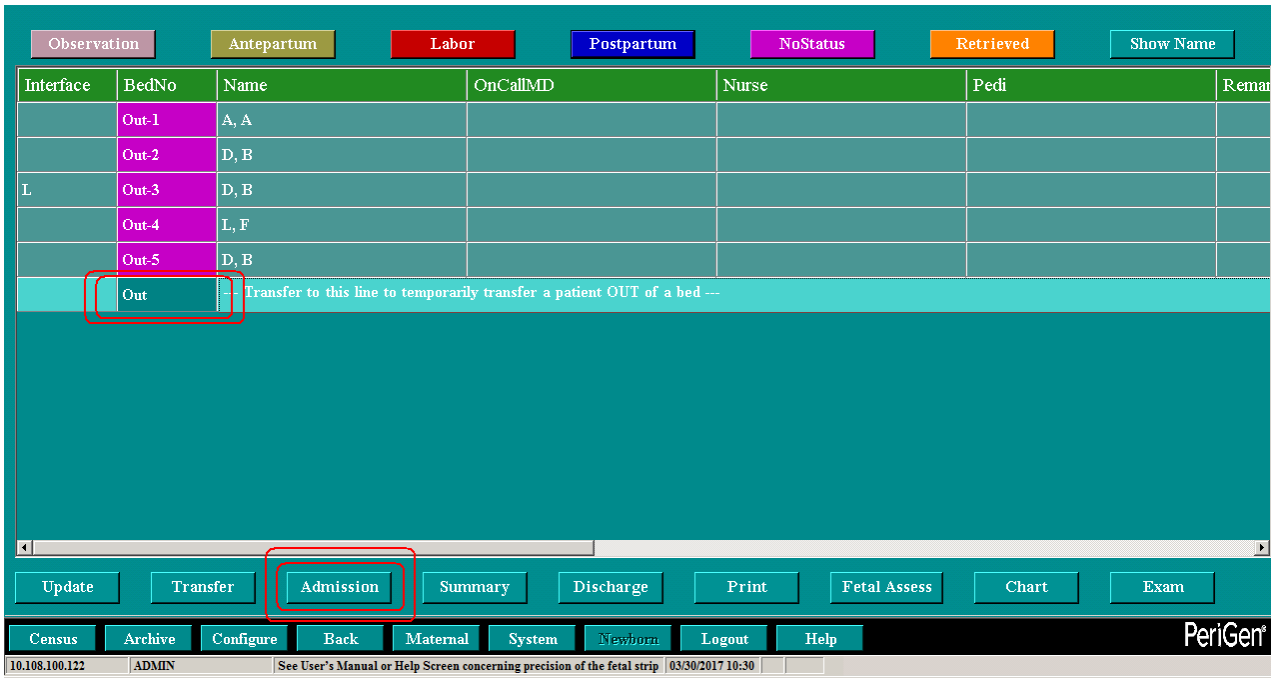
The following message was run through the interface engine into PWT (message formatted for legibility):

```
MSH|^~\\&|SendApp|SendFac|RecvApp|RecvFac|201101010001||ADT^A08|ADT201947277|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202||SIGN|SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
```

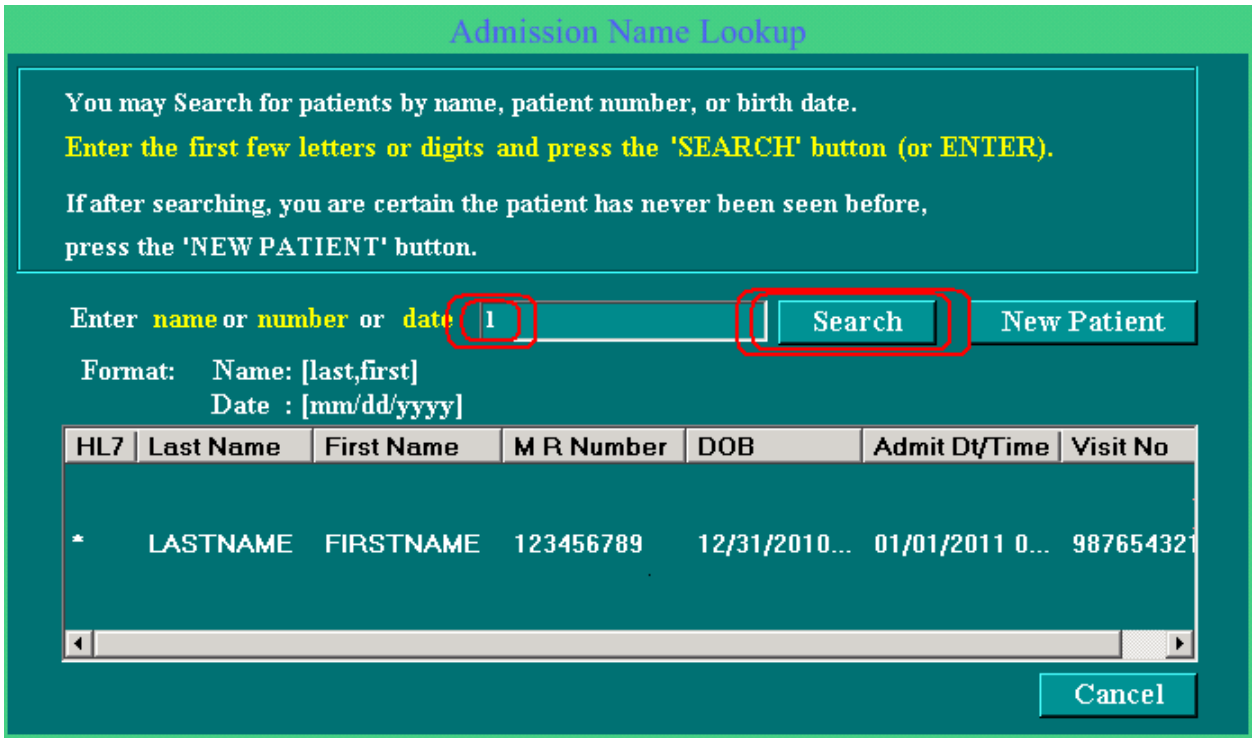
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)2020203^^^^^202^2020204
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
SERVICE||||ADMITSOURCE (e.g., ED)||||004^DOCTOR^ADMITTING|

To review the application's consumption of this message:

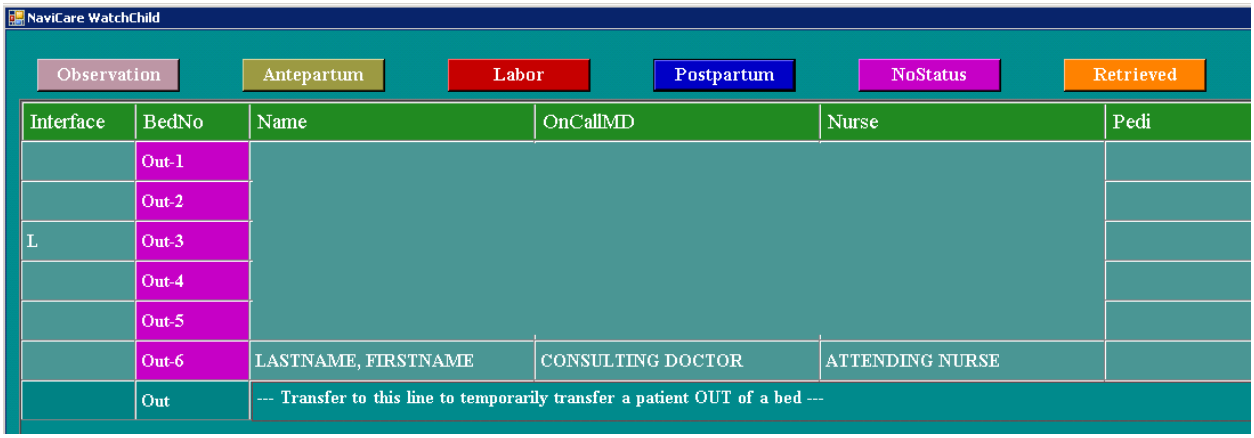
- Open the PWT application,
- Sign in, and
- Navigate to the census screen:



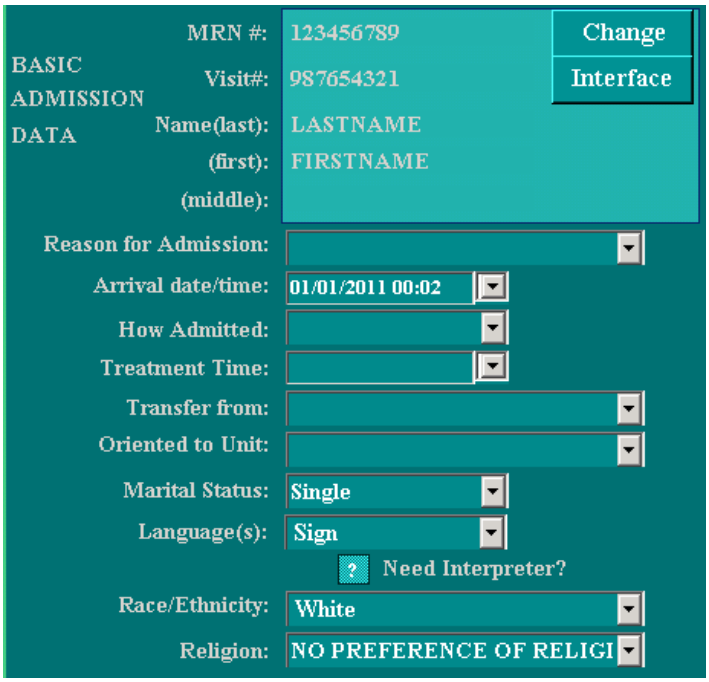
- Select a bed into which you can admit the patient, and click on "Admission" (as shown above).



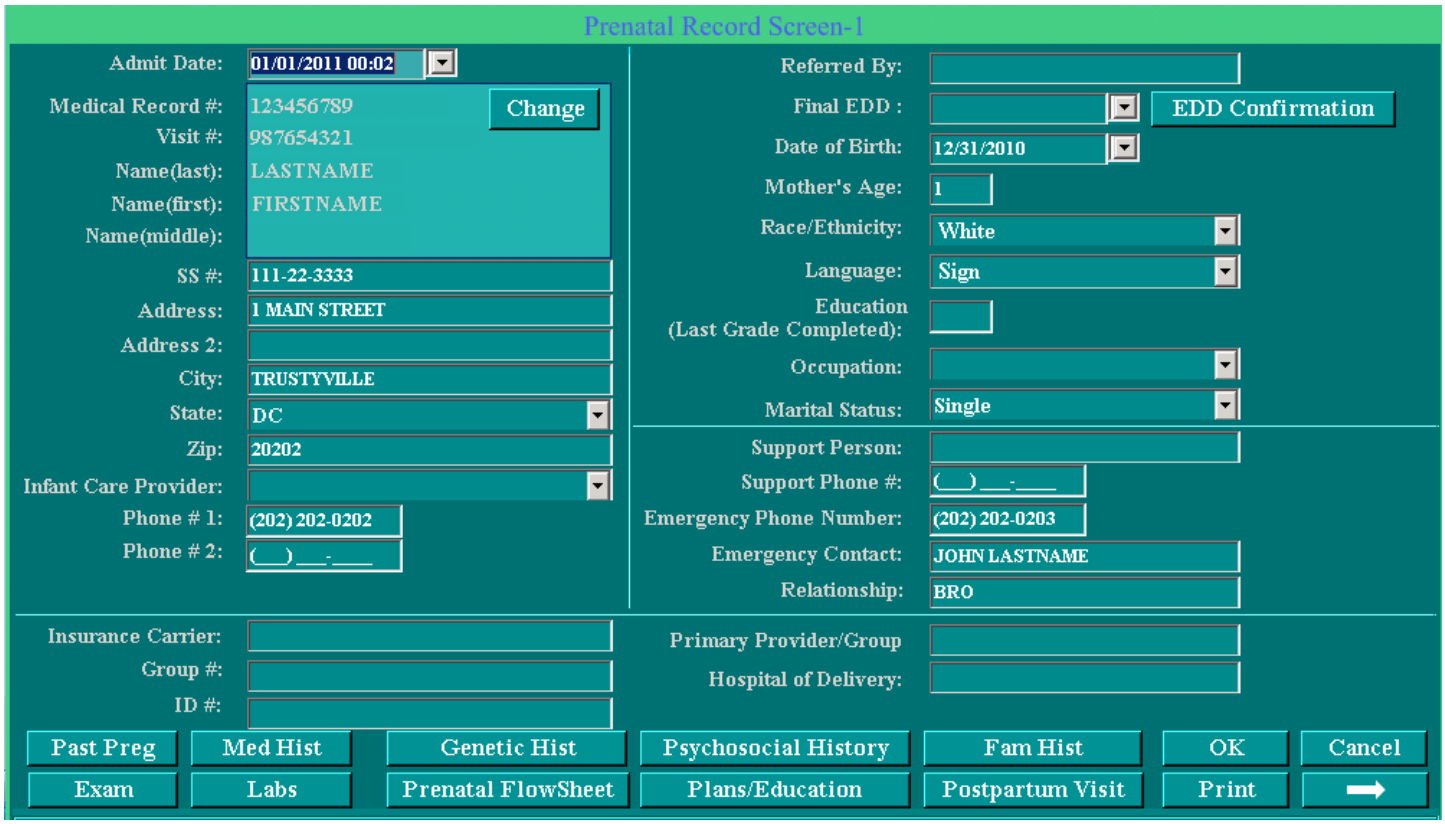
- Search for your patient based on some known value (above, searching for “1” as part of the Name, MRN, or DateTime)
- Click “Search”
- In the resulting list, patients whose information has been obtained from inbound HL7 ADT messages will have an * in the “HL7” column, as above. Select the desired patient.



- The census screen will now have your patient listed, as shown in the screen clip above. Selecting that patient and clicking on “Admission” will open the following screen:



- Notice the fields populated in the screen above; comparing them against the HL7 message at the top of this document, the data source and destination are apparent.
- Further data is found elsewhere; select again the patient from the census and click “Chart”
- Selecting “Prenatal“ reveals the following screen with imported data:



The majority of the Obstetric Admitting and Prenatal records can be populated via the HL7 interface. The above describes the "standard" capabilities of the interface, if additional pre-population is desired please contact your interfaces representative or PWT project manager.

ORU^R01	Unsolicited Results
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
{	
OBR	Observation Request
{	
OBX	Observation / Result
[{ **NTE** }]	Notes and Comments
[**SPM**]	Specimen
}	
}	

Laboratory Results

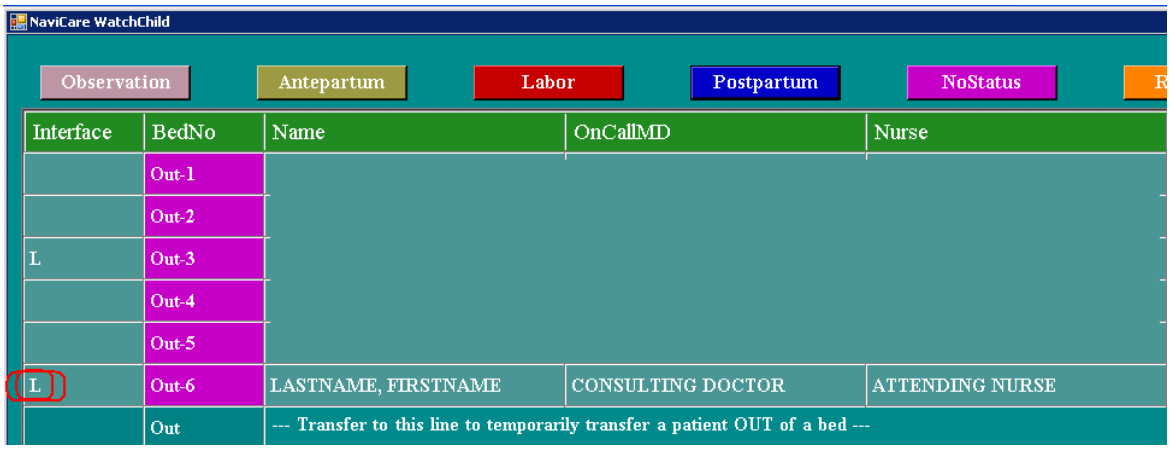
Results are received in unsolicited observation (ORU^R01) messages from LIS systems. This message type is reused for subsequent interface functionality; absent any requisite (and later-described) distinguishing information, received ORU^R01 messages are presumed to be laboratory observations/results.

Lab Example

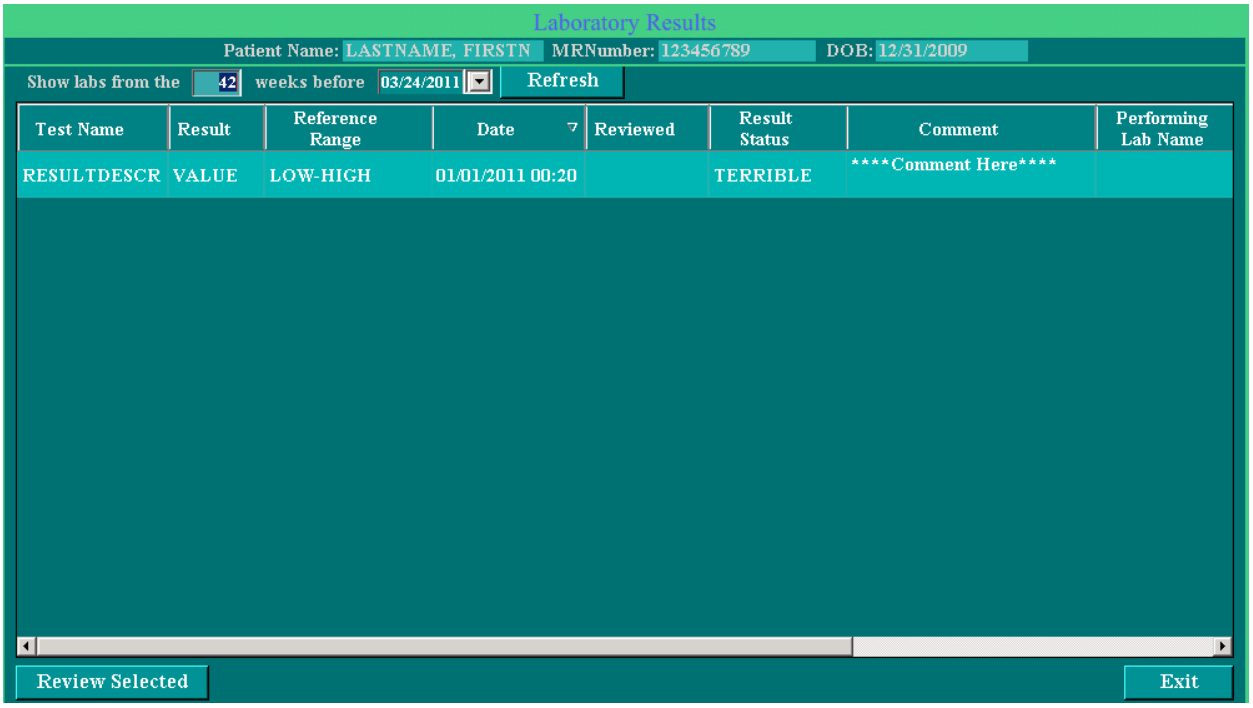
The following message was run through the interface engine into PWT (message formatted for legibility):

```
MSH|^~\\&|SendApp2|SendFac2|RecvApp|RecvFac|201101010021||ORU^R01|ADT201947278|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202|||SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)202-0203^^^^^202^2020204|
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|OUT/INPATIENT|AlternateVisitNo||||||
   |||||||||||||||||201101010002|||||||005^NURSE^ATTENDING
OBR|1|ORDER_NO1|FILLERNO1^^|TESTCODE1^TESTDESCR^^^^|||201101010020||||||201101010021|||||||
   201101010022||||||
OBX|1|ST|RESULT_CODE^RESULTDESCR^||VALUE|MG/DL|LOW-HIGH|TERRIBLE||||||||
NTE|1||****Comment Here****|
```

The PID segment of the passed message must match the MRN, first and last name, and date-of-birth of a patient who has either been manually admitted into PWT or for whom an ADT message exists and has been or will be sent to PWT. Upon arrival of a lab result message for an admitted patient, the census will show an 'L' next to the patient in the "Interface" column:



- Click on this "L" to show the Laboratory Results screen:



Pharmacy Orders

RDE^O11	Encoded Order Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXE	Pharmacy Encoded Order
{ **RXR** }	Pharmacy/Treatment Route

Pharmacy Dispenses

RDS^O13	Dispense Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXD	Pharmacy/Treatment Dispense
{ **RXR** }	Pharmacy/Treatment Route

Pharmacy Administrations

RAS^O17	Administration Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXE	Pharmacy Encoded Order
{ **RXR** }	Pharmacy/Treatment Route

Inbound Annotations
Inbound annotations are expected according to the previously described ORU^R01 message container.

Annotations themselves are to be passed in OBX segments (which may repeat). An annotation's unique identifiers are the patient's medical record number, the datetime of the annotation (with resolution fixed at "second", 120100.000 and 120100.999 would be interpreted as occurring at the same time), and the field prefix.

OBX|n||ANNOTATION EXAMPLE||SOME TEXT HERE.|||||F|||20121212121202||USER

…becomes a new annotation by user at 12:12:02 12Dec12 – "ANNOTATION EXAMPLE: SOME TEXT HERE."

DateTime of the annotation must be localtime of the PWT server, as we perform an internal conversion to UTC based on the PWT application's server time.

Inbound Annotation OBX Segment definition (all fields are required):

Outbound Vitals and Annotations
Outbound vitals and annotations are transmitted utilizing the previously described ORU^R01 container. OBR.4.2 will contain either "Vitals" or "Annotation", to differentiate the message contents.

For outbound vitals, OBX.3 will contain the vital sign being detailed; OBX.5 will hold the value and OBX.6 the unit – e.g.,

OBX|2|NM|NIBPSystolic||118|^mmHg|||||

Available vital signs include:

- Maternal Heart Rate
- SpO_2
- Maternal Mean, Systolic, and Diastolic Blood Pressure

Outbound annotations are similarly formatted, with OBX.3 equaling "Annotation" for all, and OBX.5 containing the annotation – e.g.,

OBX_n			Observation / Result
.3.1	Field Prefix	**.14**	Date/Time of the Annotation
.5$_0$	Annotation Text	**.16$_0$.1**	Responsible Observer ID
.11	Insert Mode { F = New, C = Update, D = Delete }		

OBX||TX|Annotation||Strip evaluated by RN.

Please see Appendix I for list of Annotations transmitted from PWT.

Outbound Dynamic Data Distribution (3D) Report Metadata

Outbound dynamic data distribution metadata messages are transmitted utilizing the previously described ORU^R01 container.
Please see **Appendix I** for list of Uterine Fetal Assessment, Cervical Exam transmitted from PWT

Segment and Field Definitions

The following tables describe fields directly consumed by PWT. The intention of this section is to describe which fields are actually and directly mapped into (or out of) the PWT application. Fields required by the HL7 standard specification are still required for a message to be considered valid: omission of a field otherwise required by the HL7 standard specification from tables below indicates only that it is otherwise ignored by the PWT interface engine logic.

Unless otherwise noted, any additional subfield breakdowns will default to the first segment (e.g., if XXX.3.2 is noted in the table below but a given message defines subfields within XXX.3.2, XXX.3.2.1 will be utilized by default). If a field may repeat per specification, this field will be notated within the table by either a subscript **0** or **n;** indicating our use of only the first of the repeats and ignoring all subsequent fields or processing all passed repeats, respectively. Thus $AL1_n.5_0$ indicates that all passed allergy segments are processed, but that only the first .5 segment is considered for each AL1 – any subsequent .5 segments in a given AL1 message would be ignored.

All fields are considered of type **String** save **DateTime** fields, which are considered strings but parsed into MM/DD/YYYY HH:MM:SS.

$AL1_n$			**Allergy Information**
.2	Type	**.3.2**	Code, Mnemonic, Description (Text)
.3.1	Code, Mnemonic, Description (Identifier)	**$.5_0$**	Reaction
$DG1_0$			**Diagnosis**
.3	Code	**.4**	Description
MSH			**Message Header**
.3	Sending Application	**.9**	Message Type and Trigger
.4	Sending Facility	**.10**	Control ID
.5	Receiving Application	**.11**	Processing ID
.6	Receiving Facility	**.12**	Version ID
.7	Date/Time of Message		
$NK1_0$			**Next-of-Kin**
.2	Contact Name	**$.5_0.1$**	Phone Number
.3	Relationship		
NTE			**Notes and Comments**

.1	Set ID	**.3$_n$**	Comment
OBR			**Observation Request**
.3	Filler Order Number	**.4.2**	Universal Service ID (Text)
.4.1	Universal Service ID (Identifier)	**.7**	Observation Date/Time
OBX$_n$			**Observation / Result**
.3.1	Observation Identifier (Identifier)	**.11**	Observation Result Status { P, F, C }
.3.2	Observation Identifier (Text)	**.23**	Performing Organization's Name
.3.4	Observation Identifier (Alternate ID)	**.24.1**	Performing Organization's Street Address
.5$_n$	Observation Value	**.24.3**	Performing Organization's City
.6.2	Units (Text)	**.24.4**	Performing Organization's State
.7	References Range	**.24.5**	Performing Organization's Zip Code
.8$_0$	Abnormal Flags		
ORC			**Common Order**
.1	Order Control	**.7.2.2**	Quantity Timing, Explicit Time Interval
.3	Filler Order Number	**.9**	Date/Time of Transaction

PID			**Patient Identification**
.3$_0$.1	Internal Patient ID (Medical Record Number)	**.11$_0$.4**	Patient's Address – State
.5$_0$.1	Family Name	**.11$_0$.5**	Patient's Address – Zip Code
.5$_0$.2	Given Name	**.13$_0$.1**	Home Phone Number
.5$_0$.3	Middle Name or Initial	**.15.1**	Primary Language (Identifier)
.7	Date/Time of Birth	**.16.1**	Marital Status (Identifier)
.10$_0$.1	Race (Identifier)	**.17.1**	Religion (Identifier)
.11$_0$.1	Patient's Address – Street Address	**.18.1**	Patient Account / Visit Number
.11$_0$.3	Patient's Address – City	**.21$_0$.1**	Mother's Medical Record Number
PV1			**Patient Visit**
.3.3	Assigned Patient Location (Bed)	**.17.3**	Admitting Doctor's Given Name
.9.2	Consulting Doctor's Family Name	**.44**	Admission Date/Time
.9.3	Consulting Doctor's Given Name	**.52.2**	Attending Nurse's Family Name
.17.2	Admitting Doctor's Family Name	**.52.3**	Attending Nurse's Given Name
PV2			**Patient Visit – Additional Information**
.3.2	Reason for Admission		
RXD			**Pharmacy/Treatment Dispense**
.2.1	Dispense Give Code (Identifier)	**.4**	Actual Dispense Amount
.2.2	Dispense Give Code (Text)	**.5.2**	Actual Dispense Units (Text)
.3	Date/Time Dispensed		
RXE			**Pharmacy/Treatment Encoded Order**
.1.4	Quantity/Timing, Start Date/Time	**.3**	Minimum Give Amount

.2.1	Give Code (Identifier)	**.5.2**	Give Units (Text)
.2.2	Give Code (Text)		
RXR			**Pharmacy/Treatment Route**
.1.2	Route (Text)		
SPM			**Specimen**
.8.2	Source Site (Text)	**.24.2**	Condition (Text)

3 Newborn Interface Specification

The purpose of this document is to outline the PeriWatch Tracings (**PWT**) expectations for HL7 message exchange with other systems. This document should form the foundation for interface design discussions, and presumes prior knowledge of the Health Level Seven (**HL7**) messaging standard; details available at http://www.hl7.org.

3.1 Communications

The PWT integrations engine resides on PWT training and production application servers, there sharing server IPs, network domain authorization/restrictions, and server hardware and database with the related application. As hardware and database are shared between application and integrations engine, PWT expresses strong preference towards the configuration of a **client-side filter**, where possible only passing messages related to existing or potential PWT patients to our engine for consideration and consumption. Often this filtering is accomplished by logic parsing PV1.3.1, passing messages only for patients with an assigned location of (e.g.) LDRP, NICU; specific implementation of these filters is at client discretion. Additional requisite filtering may be required; details below where relevant.

Interface messaging occurs across TCP/IP connections. Our standard implementation port for incoming messages is **17676**.

Splitting of inbound message types occurs within the engine, and therefore messages should be collected at client systems and transmitted to PWT in a unitary stream where multiple systems are involved (e.g., collect ADT and Lab messages for serial transmission to PWT on port 17676). Due to the loosely-coupled nature of the integrations engine with the application, messages need not be properly sequenced to be properly processed (e.g., reception of a lab message for a patient for which we have not yet received an ADT message is acceptable; reception of an ADT update message prior to reception of an ADT admit message is acceptable).

PWT is configured for serialized/bimodal message handling, transmitting or waiting for reception of acknowledgment messages immediately after message reception/transmission. Failed transmissions will retry or wait for retry until five attempts have passed, then closing the port and marking the route in error.

The low level format of the HL7 data stream adheres to standard:

<SB>dddd**<EB><CR>**

…where **<SB>** is the start block character (1 byte ASCII <VT>, 0x0B),
dddd represents the HL7 message data stream,

<EB> is the end block character (1 byte ASCII <FS>, 0x1C), and
<CR> is the carriage return character (0x0D).

3.2 Charting Interface Description

Where a typical downstream ancillary will react to the reception of an ADT^A01 message by admitting that patient to their census, PWT will not take any action – instead the message contents are placed in a pool of data from which a user can draw information during a manual admission into the application. As this is a deviation from common expectation, feel free to ask for more detail.

3.3 HL7 Message Definitions

The PWT interface standard adheres to the published HL7 standards for any v2.x message type through v2.6. Valid transmitted message segments which are not mentioned in the following message definitions are ignored by our application.

Message delimiters adhere to standard: |,^,&,~, and \ being reserved characters to delimit fields, components, subcomponents, repeating fields, and escaped characters (respectively).

PeriWatch Newborn currently has the capacity to process inbound HL7 messages pertaining to registration (ADT) and lab results (ORU). Provision of additional inbound or outbound message exchange is outside the scope of this document.

The following sections define the required and optional components of PWT HL7 messages. Indentations denote logical message sub-blocks; fields in curly braces (**{}**) may repeat, those in square brackets (**[]**) are optional per specification.

3.4 Message Structures

ADT^A0{1,4,5,8}	Admit, Registration, PreAdmit, Update
MSH	Message Header
EVN	Event Type
PID	Patient Identifier
NK1	Next-of-Kin / Emergency Contact
PV1	Patient Visit Information
[{ **PV2** }]	Additional Patient Visit Information
[{ **OBX** }]	Observation Segments
[{ **AL1** }]	Patient Allergy Information
[{ **DG1** }]	Patient Diagnoses

Admit, Discharge, Transfer (ADT)
PWT only processes received Admit, Registration, Pre-Admit, and Update (A01, A04, A05, and A08, respectively) messages; all others are ignored and should be filtered from transmission. PWT does not transmit any ADT messages.

3.5 ADT Example

The following message was run through the interface engine into PWT (message formatted for legibility):

```
MSH|^~\\&|SendApp|SendFac|RecvApp|RecvFac|201101010001||ADT^A08|ADT201947277|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202||SIGN|SINGLE|NO PREFERENCE OF RELIGION|987654321|111-22-
   3333||||||||||
```

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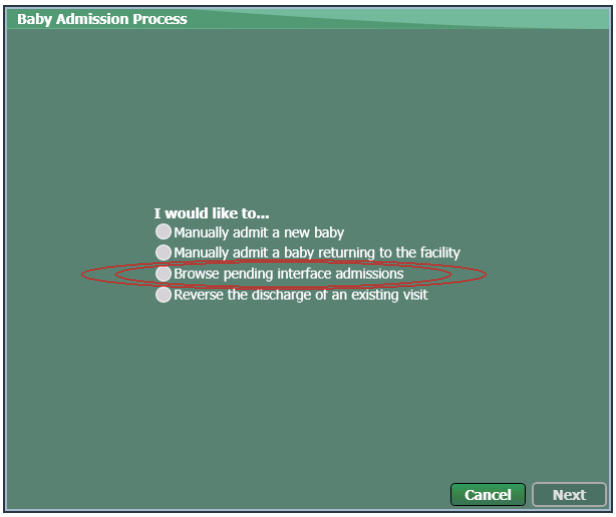
Cary, North Carolina 27518
Phone: 855-236-4353
Fax: 877-327-0084

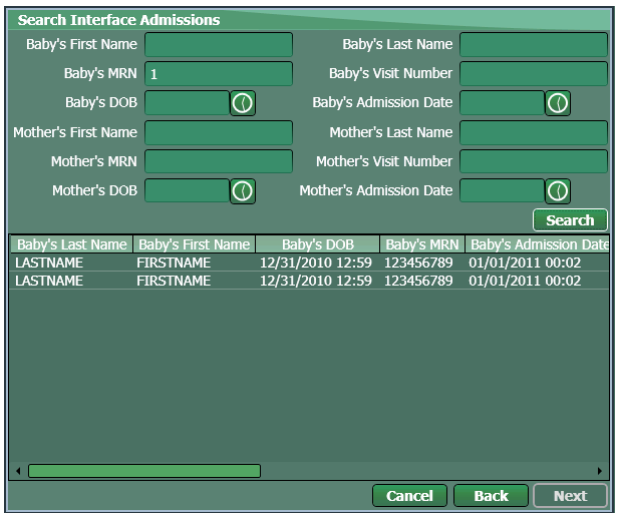
```
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)2020203^^^^^202^2020204
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
SERVICE|||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|
OUT/INPATIENT|AlternateVisitNo||||||||||||||||||||||||||201101010002|||||||005^NURSE^ATTENDING
```

To review the application's consumption of this message:

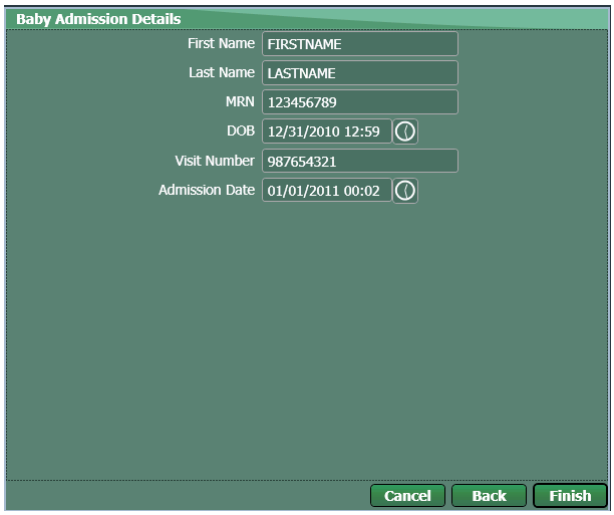
- Open the PeriWatch Newborn application,
- Sign in, and
- Navigate to the census screen.
- Select "Admit" from the lower-left, then "Browse pending interface admissions".



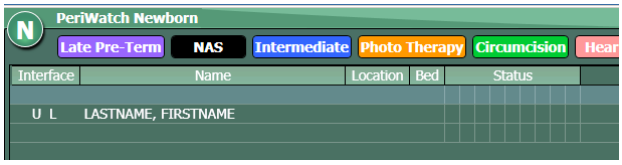
- Search for your patient based on some known value (above, searching for "1" as part of the Name, MRN, or DateTime); if you do not want to constrain the search by Baby's Admission Date, you must select the text in the field and hit "Delete" or "Backspace" to clear the contents.
- Click "Search"



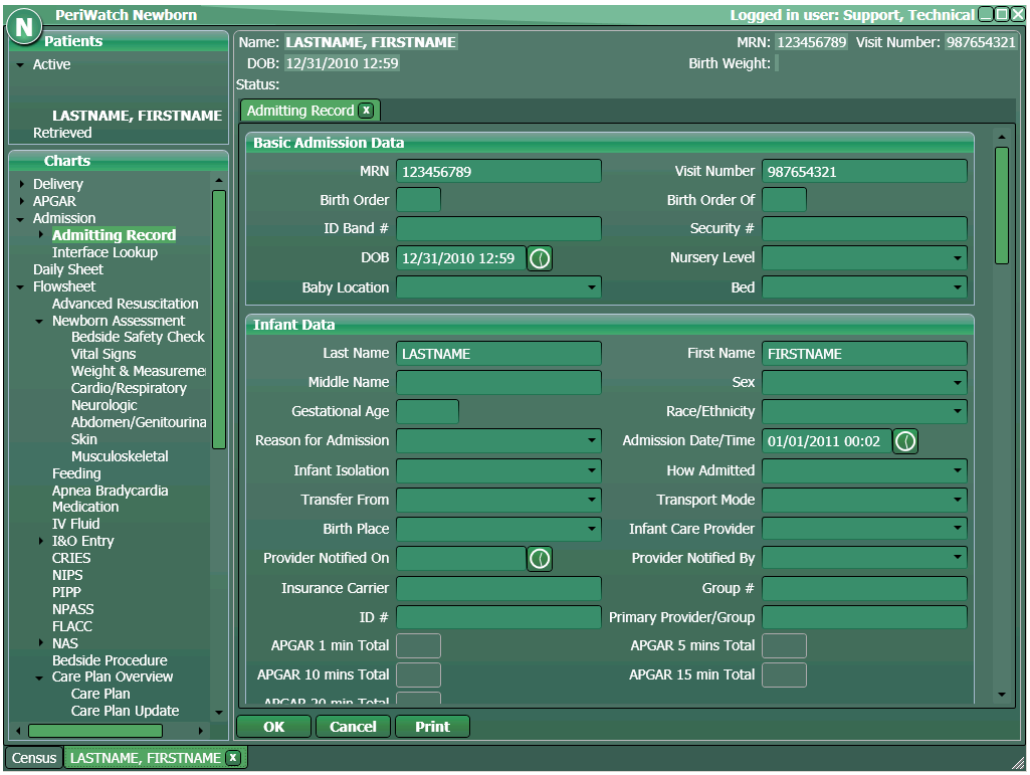
- In the resulting list, patients whose information has been obtained from inbound HL7 ADT messages will be returned. You will see one line in the table for each message received: unlike the maternal application, a single patient may have many results returned here. Selection of the top-most entry will generally return the most up-to-date and complete message. Select the desired patient.



- Verify the retrieved information and click Finish when complete.



- The census screen will now have your patient listed, as shown in the screen clip above. Selecting that patient and clicking on “Chart”, then “Admission” will open the following screen:



- Notice the fields populated in the screen above; comparing them against the HL7 message at the top of this document, the data source and destination are apparent.
- Where new ADT messages have arrived for a patient, click on the "U" in the census next to their name. On the resulting search screen, uncheck Admit Date Time and Date of Birth search options and click "Search", to review new messages for incorporation into the patient record.

ORU^R01	Unsolicited Results
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
{	
OBR	Observation Request
{	
OBX	Observation / Result
[{ **NTE** }]	Notes and Comments
[**SPM**]	Specimen
}	
}	

Laboratory Results

Results are received in unsolicited observation (ORU^R01) messages from LIS systems. This message type is reused for subsequent interface functionality; absent any requisite (and later-described) distinguishing information, received ORU^R01 messages are presumed to be laboratory observations/results.

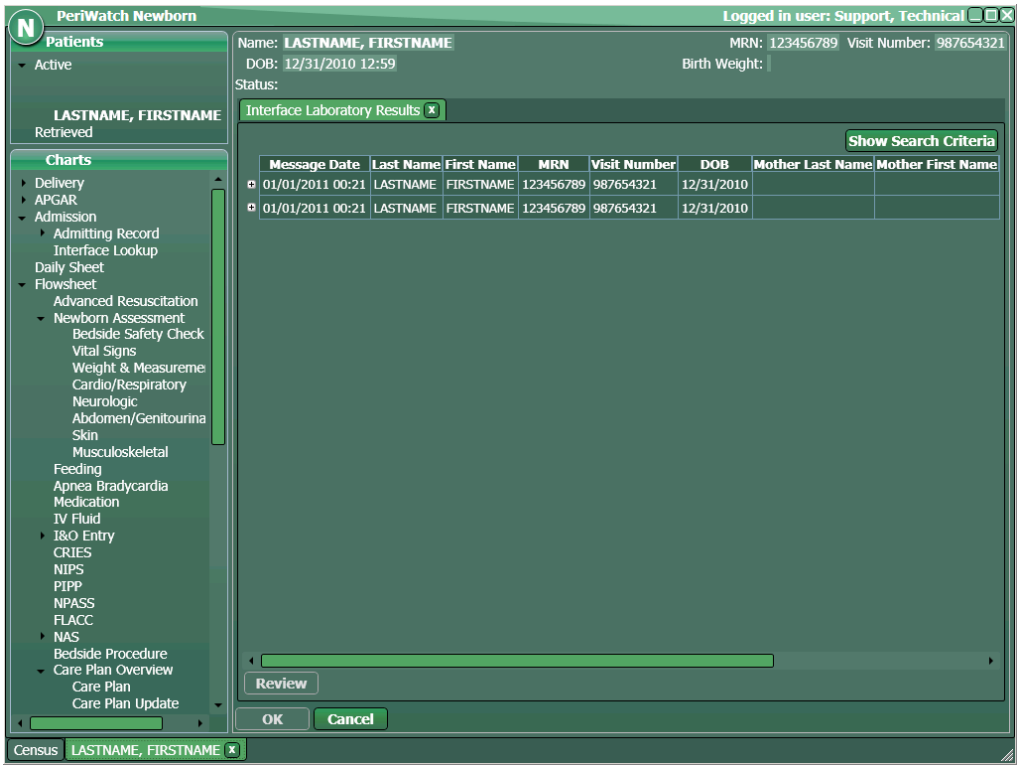
3.6 Lab Example

The following message was run through the interface engine into PWT (message formatted for legibility):

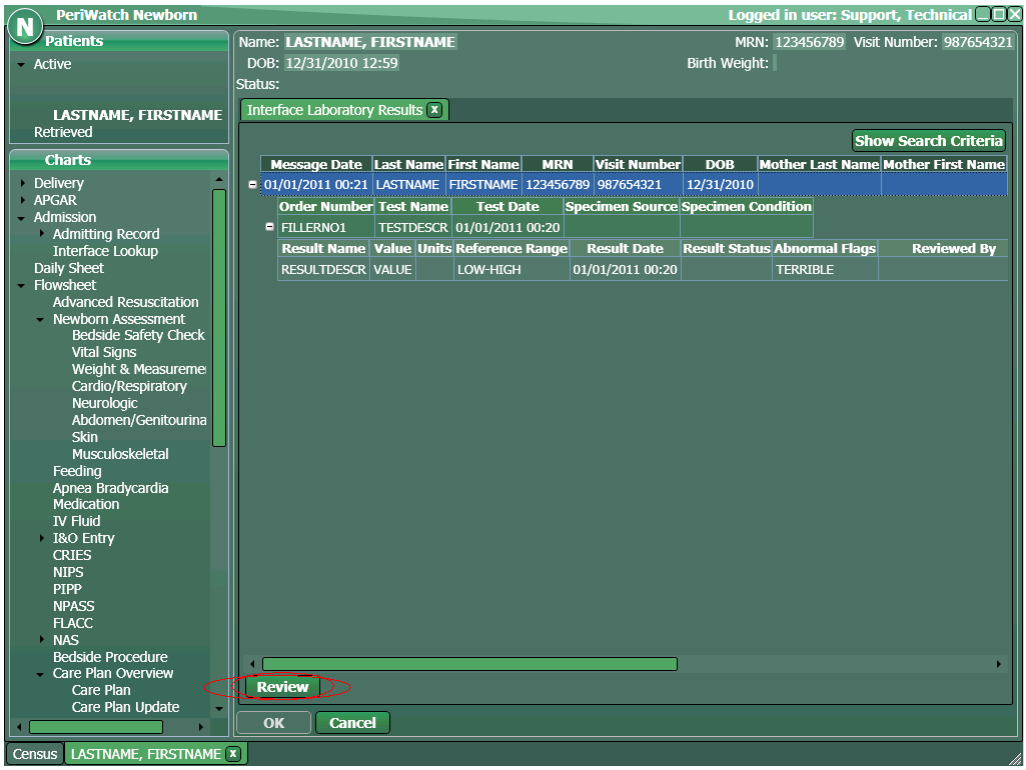
```
MSH|^~\\&|SendApp2|SendFac2|RecvApp|RecvFac|201101010021||ORU^R01|ADT201947278|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202|||SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)202-0203^^^^^202^2020204|
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|OUT/INPATIENT|AlternateVisitNo||||||
   |||||||||||||||||201101010002|||||||005^NURSE^ATTENDING
OBR|1|ORDER_NO1|FILLERNO1^^|TESTCODE1^TESTDESCR^^^^|||201101010020||||||201101010021|||||||
   201101010022||||||
OBX|1|ST|RESULT_CODE^RESULTDESCR^||VALUE|MG/DL|LOW-HIGH|TERRIBLE|||||||||
NTE|1||****Comment Here****|
```

The PID segment of the passed message must match the MRN, first and last name, and date-of-birth of a patient who has either been manually admitted into PWT or for whom an ADT message exists and has been or will be sent to PWT. Upon arrival of a lab result message for an admitted patient, the census will show an 'L' next to the patient in the "Interface" column.

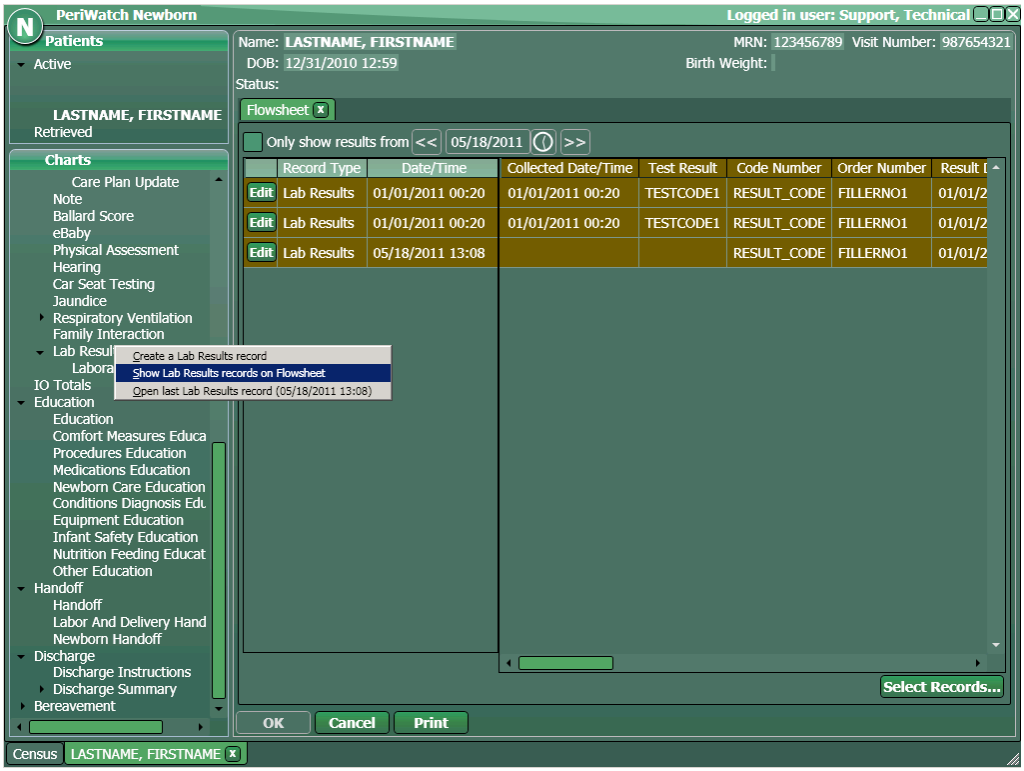
- Click on this "L" to show the Laboratory Results screen:



- There is a line for each inbound laboratory message – each message may contain one or many actual results. Expand the entries by clicking the ⊞ to see the results themselves; click on the message row and select "Review" to add the lab result to the patient's chart. (Lab messages for the patient will not be added to the patient's medical record within PeriWatch Newborn unless reviewed.)



- To view results which have been reviewed and posted on the chart, right-click on "Lab Results" and select "Show Lab Results records on Flowsheet"


PeriWatch Newborn
Logged in user: Support, Technical
Patients
Active
LASTNAME, FIRSTNAME
Retrieved
Charts
Care Plan Update
Note
Ballard Score
eBaby
Physical Assessment
Hearing
Car Seat Testing
Jaundice
Respiratory Ventilation
Family Interaction
Lab Resul
Labora
IO Totals
Education
Education
Comfort Measures Educa
Procedures Education
Medications Education
Newborn Care Education
Conditions Diagnosis Edu
Equipment Education
Infant Safety Education
Nutrition Feeding Educat
Other Education
Handoff
Handoff
Labor And Delivery Hand
Newborn Handoff
Discharge
Discharge Instructions
Discharge Summary
Bereavement
Create a Lab Results record
Show Lab Results records on Flowsheet
Open last Lab Results record (05/18/2011 13:08)
Name: LASTNAME, FIRSTNAME
MRN: 123456789 Visit Number: 987654321
DOB: 12/31/2010 12:59
Birth Weight:
Status:
Flowsheet
Only show results from << 05/18/2011 >>
Record Type
Date/Time
Collected Date/Time
Test Result
Code Number
Order Number
Edit Lab Results 01/01/2011 00:20 01/01/2011 00:20 TESTCODE1 RESULT_CODE FILLERNO1 01/01/2
Edit Lab Results 01/01/2011 00:20 01/01/2011 00:20 TESTCODE1 RESULT_CODE FILLERNO1 01/01/2
Edit Lab Results 05/18/2011 13:08 RESULT_CODE FILLERNO1 01/01/2
Select Records...
OK
Cancel
Print
Census
LASTNAME, FIRSTNAME

3.7 Segment and Field Definitions

The following tables describe fields directly consumed by PWT. The intention of this section is to describe which fields are actually and directly mapped into (or out of) the PWT application. Fields required by the HL7 standard specification are still required for a message to be considered valid: omission of a field otherwise required by the HL7 standard specification from tables below indicates only that it is otherwise ignored by the PWT interface engine logic.

Unless otherwise noted, any additional subfield breakdowns will default to the first segment (e.g., if XXX.3.2 is noted in the table below but a given message defines subfields within XXX.3.2, XXX.3.2.1 will be utilized by default). If a field may repeat per specification, this field will be notated within the table by either a subscript **0** or **n;** indicating our use of only the first of the repeats and ignoring all subsequent fields or processing all passed repeats, respectively. Thus $AL1_n.5_0$ indicates that all passed allergy segments are processed, but that only the first .5 segment is considered for each AL1 – any subsequent .5 segments in a given AL1 message would be ignored.

All fields are considered of type **String** save **DateTime** fields, which are considered strings but parsed into the XML-defined UTC DateTime format (with GMT-offset presumed equal to that of the host system).

Concerning the capture of an infant's mother's information: $PID.21_0.1$ (Mother's MRN) is populated for infants where available. Where NK1.3 (Relationship) is "MTH" or "MOTHER", the information from this segment is used to prepopulate information regarding the infant's mother. Currently First and Last Name are the only captured values; additional data capture may be possible upon request.

$AL1_n$			**Allergy Information**
.2	Type	**.3.2**	Code, Mnemonic, Description (Text)
.3.1	Code, Mnemonic, Description (Identifier)	**$.5_0$**	Reaction
$DG1_0$			**Diagnosis**
.3	Code	**.4**	Description
MSH			**Message Header**
.3	Sending Application	**.9**	Message Type and Trigger
.4	Sending Facility	**.10**	Control ID
.5	Receiving Application	**.11**	Processing ID
.6	Receiving Facility	**.12**	Version ID
.7	Date/Time of Message		
$NK1_0$			**Next-of-Kin**
.2	Contact Name	**$.5_0.1$**	Phone Number
.3	Relationship		
NTE			**Notes and Comments**
.1	Set ID	**$.3_n$**	Comment
OBR			**Observation Request**
.3	Filler Order Number	**.4.2**	Universal Service ID (Text)
.4.1	Universal Service ID (Identifier)	**.7**	Observation Date/Time
OBX_n			**Observation / Result**
.3.1	Observation Identifier (Identifier)	**.11**	Observation Result Status { P, F, C }
.3.2	Observation Identifier (Text)	**.23**	Performing Organization's Name
.3.4	Observation Identifier (Alternate ID)	**.24.1**	Performing Organization's Street Address

.5$_n$	Observation Value	**.24.3**	Performing Organization's City
.6.2	Units (Text)	**.24.4**	Performing Organization's State
.7	References Range	**.24.5**	Performing Organization's Zip Code
.8$_0$	Abnormal Flags		
ORC			**Common Order**
.1	Order Control	**.7.2.2**	Quantity Timing, Explicit Time Interval
.3	Filler Order Number	**.9**	Date/Time of Transaction

PID			**Patient Identification**
.3.1	Internal Patient ID (Medical Record Number)	**.11.4**	Patient's Address – State
.5.1	Family Name	**.11.5**	Patient's Address – Zip Code
.5.2	Given Name	**.13$_0$.1**	Home Phone Number
.5.3	Middle Name or Initial	**.15.1**	Primary Language (Identifier)
.7	Date/Time of Birth	**.16.1**	Marital Status (Identifier)
.10$_0$.1	Race (Identifier)	**.17.1**	Religion (Identifier)
.11.1	Patient's Address – Street Address	**.18.1**	Patient Account / Visit Number
.11.3	Patient's Address – City	**.21$_0$.1**	Mother's Medical Record Number
PV1			**Patient Visit**
.3.3	Assigned Patient Location (Bed)	**.17.3**	Admitting Doctor's Given Name
.9.2	Consulting Doctor's Family Name	**.44**	Admission Date/Time
.9.3	Consulting Doctor's Given Name	**.52.2**	Attending Nurse's Family Name
.17.2	Admitting Doctor's Family Name	**.52.3**	Attending Nurse's Given Name
PV2			**Patient Visit – Additional Information**
.3.2	Reason for Admission		
SPM			**Specimen**
.8.2	Source Site (Text)	**.24.2**	Condition (Text)

4 Appendix I

4.1 Strip Annotations:

Field	Value
Intrauterine resuscitation	Position changed IV bolus initiated- See MAR Oxygen at 10L/min via non-rebreather mask Called for assistance Pitocin off- See MAR Tocolytics given- See MAR Ultrasound to verify fetal heart rate
Fetal interventions	Scalp stimulation between contractions Ultrasound to verify fetal heart rate RN continuously at bedside evaluating FHR Modified pushing effort Vibroacoustic stimulation Amnioinfusion started Amnioinfusion stopped Amnioinfusion fluid returned
Membrane	Intact SROM- See EMR AROM- See EMR Bulging
Monitor	EFM applied Toco applied EFM adjusted Toco adjusted FSE applied FSE discontinued FHR via doppler FHR via auscultation Telemetry/wireless applied IUPC inserted IUPC zeroed IUPC flushed IUPC discontinued

Field	Value
I&O Interventions	Ice chips given-Document volume in I/O Popsicle given-Document volume in I/O Clear liquids given-Document volume in I/O Up to bathroom Bedpan given- Document volume in I/O Straight catheter- Document volume in I/O PeriCare given Foley catheter inserted- Document volume in I/O Foley catheter discontinued- Document volume in I/O
IV	IV started Labs drawn with IV start IV bolus started- See MAR IV bolus stopped- See MAR New IV bag- See MAR IV saline locked IV fluids discontinued- See MAR
Induction/Augmentation	Pitocin started- See MAR Pitocin increased- See MAR Pitocin decreased- See MAR Pitocin discontinued- See MAR Cervical ripening agent- See MAR Laminaria Cervical ripening balloon placed Cervical ripening balloon discontinued
Positions	Birthing ball Chair Hands/knees High Fowlers Jet Hydrotherapy Knee chest Left lateral Left tilt Low Fowlers Peanut ball Right lateral Right tilt Rocking chair

Field	Value
	Semi-fowlers Shower Sitting Squatting Standing Supine Tailor sitting Trendelenburg Walking Other
Comfort Measures	Underpad changed Coached with breathing/relaxation techniques Up to shower Assist to tub Laboring in tub Back rub given Hot pack applied Cold pack applied Family support at bedside Aromatherapy in use Doula support at bedside Hypnotherapy
Medication	Terbutaline given- See MAR Ephedrine given- See MAR Medication given- See MAR Magnesium sulfate bolus started- See MAR Magnesium sulfate bolus complete- See MAR Magnesium sulfate maintenance dose started- See MAR Magnesium sulfate maintenance dose discontinued- See MAR
Pushing Preparation	Patient reports feeling pressure Foley removed- Document volume in I/O Family at bedside Doula at bedside

Field	Value
	RN continually at bedside evaluating FHR Provider at bedside to push with patient Leg support by family/significant other Patient instructed in pushing technique
Second stage position	Semi-fowlers High fowlers Right lateral Left lateral Squatting Hands/knees Knee chest Squatting bar In stirrups Pushing on side with leg up in stirrup Birthing bar Support of significant other Pushing with sheet around birthing bar
Pushing Effectiveness	Patient pushing well Patient pushing ineffectively Patient unable to feel pressure to push Baby crowning Making slow progress Epidural pump off
Delivery Preparation	Provider called to room for delivery Patient set up for delivery Perineal prep done Provider at bedside for delivery Second RN/OB Tech called for delivery NICU called for delivery NICU at bedside Anesthesia called for perineal dose Anesthesia at bedside
Labor provider name/title	<Free Text>
Method of notification	Electronic page Text page

Field	Value
	Overhead page Provider phoned Left message Provider at bedside
Request	Evaluate remote Evaluate in person Attend delivery
Notification Reason	Decels Fetal baseline change Variabiliity change Membrane status Labor status Uterine activity Pain Maternal vital signs change Bleeding Lab/diagnostic study Patient request Status update
Notification Reason	Decels Fetal baseline change Variabiliity change Membrane status Labor status Uterine activity Pain Maternal vital signs change Bleeding Lab/diagnostic study Patient request Status update
Response	Provider returned call Provider given report Provider to bedside Provider leaves bedside Provider in department Provider reviewed fetal monitor strip

Field	Value
Strip Annotation (free text)	<Free Text>

4.2 Uterine Fetal Assessment:

Field	Value
Monitor	External Toco Internal IUPC Palpation None Other
Frequency	<Numeric><Text> Minutes
Contractions per 10 minutes	<Numeric><Text> Minutes
Duration	<Numeric><Text> Minutes
Intensity	Mild Mild/Moderate Moderate Moderate/Strong Strong Other
Intensity IUPC mmhg	<Numeric><Text>
Resting Tone	Mild Moderate Strong Soft Rigid Other
MVU	<Numeric><Text>
Characteristics	Regular Irregular Irritability Tachysystole Other
Patient Position	Birthing ball Chair Hands/knees

Field	Value
	High Fowlers Jet Hydrotherapy Knee chest Left lateral Left tilt Low Fowlers Peanut ball Right lateral Right tilt Rocking chair Semi-fowlers Shower Sitting Squatting Standing Supine Tailor sitting Trendelenburg Walking Other
Comments	<Text>
Monitor	External Ultrasound Internal Fetal Spiral Electrode Auscultation Doppler Fetascope None Other
Presentation	Cephalic Cephalic Face Cephalic Brow Breech Breech Complete Breech Frank Breech Footling Shoulder Hand Compound

Field	Value
	Transverse Lie Oblique Lie Cord Unknown Other
Baseline Variability	Absent Minimal (1-5 BPM) Moderate (6-25 BPM) Marked (>25 BPM) Unable to determine
Categories	Category I Category II Category III
FHR	<Text>
FHR Rhythym	Regular Irregular
Characteristics	Bradycardia Indeterminate Normal Sinusoidal Tachycardia
Accels	Present Absent
Decels	None Early Late Variable Prolonged Present
Reoccurrence	Recurrent Intermittant
Checklist	Met Not Met
Comments	<Text>
Membrane	Intact SROM AROM

Field	Value
	PROM PPROM Prolonged Rupture of Membranes Bulging Unknown
Fluid	Clear Cloudy Bloody Meconium Stained Meconium Thin Meconium Moderate Meconium Thick Yellow Brown Green Tinted Pinkish Unknown Other
Amount	<Text>
Rupture Date/Time	<Numeric><Text>

4.3 Cervical Exam:

Field	Value
Dilitation	0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5 5.5

Field	Value
	6 6.5 7 7.5 8 8.5 9 9.5 10 0.5 Fingertip 9.5 Anterior Lip
Effacement	0 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Station	0 -1 -2 -3 -4 -5 1

Field	Value
	2 3 4 5 Crown
Fetal Position	OA LOA ROA OP LOP ROP OT LOT ROT
Cervical Position	Anterior Posterior
Consistency	Soft Medium Firm
Presentation	Cephalic Cephalic Face Cephalic Brow Breech Breech Complete Breech Frank Breech Footling Shoulder Hand Compound Transverse Lie Oblique Lie Cord Unknown
Comments	<Text>
Examined by	Physician/Nurse drop down

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5 Technical Support

PeriGen® IT Solutions Technical Support is available 24/7, 365 days a year. If you need help, please call: 1-855-236-4353.

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